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Project Title: Psycholinguistic Analysis- Do Language Models Resolve Pronoun Ambiguity Using Human-like Linguistic Cues?

Research Question 1: Do transformer language models resolve Winograd-style pronoun ambiguity using semantic reasoning similar to human cognitive processes?

Research Question 2: Which linguistic cues - semantic reasoning, syntactic structure, or lexical associations - most strongly influence model disambiguation decisions?

Hypothesis 1: Transformer-based masked language models (e.g., BERT) will achieve high baseline accuracy on pronoun resolution tasks.

Hypothesis 2: Model performance will vary depending on the linguistic cues present in the sentence.

Hypothesis 3: Sentences requiring deeper commonsense reasoning or causal inference will produce more model errors than sentences relying primarily on simpler structural or lexical cues.

Methods: This project evaluates language models as psycholinguistic subjects, using computational analysis to test hypotheses about language processing.

1. Sentence Reconstruction: Each dataset example contains a sentence with a blank and two candidate referents. For each instance, two complete sentences will be constructed by inserting each candidate referent into the blank.

2. Language Model Evaluation: Using a pretrained masked language model through a Python-based pipeline, we will compute the token-level pseudo-log-likelihood for the referent in each candidate sentence. The sentence receiving the higher probability will be taken as the model's prediction.

3. Perturbation-Based Analysis: To probe the linguistic cues influencing model decisions, controlled perturbations will be applied to a subset of sentences. These modifications may include synonym substitutions, adjective polarity changes, or structural reordering that preserves meaning. Model predictions will be evaluated before and after perturbation to observe whether small linguistic changes alter the resolution decision.

4. Error Pattern Analysis: Model errors and prediction changes under perturbations will be examined to determine whether models rely primarily on semantic reasoning or on simpler lexical associations and statistical cues.

Datasets: WinoGrande (Sub-sample): A random subset of approximately 300 sentences sampled from the WinoGrande dataset (~44,000 pronoun disambiguation problems designed to test commonsense reasoning).

Winograd Schema Challenge (WSC273): The original curated dataset of 273 Winograd schemas may be used for supplementary comparison and validation.

Key Research Paper References:

1. Sakaguchi, K., Bras, R. L., Bhagavatula, C., & Choi, Y. (2019). *WinoGrande: An Adversarial Winograd Schema Challenge at Scale*. arXiv:1907.10641.

2. Yamakoshi, M., et al. (2023). *Causal Interventions Expose Implicit Situation Models for Commonsense Language Understanding*.

3. Jurafsky, D., & Martin, J. H. (2013). *Speech and Language Processing: An Introduction to Natural Language Processing, Computational Linguistics and Speech Recognition*.